When Regression GATE Estimates Fail: Endpoint Support, Contrast Leverage, and Symmetry

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Abstract

Regression estimators of the global average treatment effect (GATE) extrapolate from a mixed treatment assignment to two global endpoints. How reliable is that extrapolation? We conduct a fixed-graph error audit rather than propose a universally superior estimator. A finite-sample decomposition separates conditional endpoint bias from covariance-weighted contrast leverage. A support counterexample shows why observed fit alone cannot certify accuracy; a complement-symmetry result explains why balanced experiments can conceal quadratic misspecification. Across 24,000 simulated experiments on controlled synthetic networks and two public topologies, larger degree increases variance at fixed size, unnecessary but correctly specified features can sharply increase error, and count-versus-fraction misspecification is consequential on heterogeneous graphs. A threshold mechanism can have essentially zero observed signal residual yet a missing global spillover effect. Correlated errors with unit marginal variance invalidate iid regression intervals; known covariance repairs this only when contrast bias is absent. Clustering improves some settings but is not uniformly beneficial. We report Monte Carlo uncertainty, estimator availability, and rare-exposure limitations, and release the full simulation and reporting code. Conclusions concern the stated response classes and fixed graphs, not arbitrary interference or real-platform effects.

1 Introduction

A platform often wants the effect of treating everyone, not the direct effect of treating one user while most neighbors remain untreated. With network interference, these are different targets. A regression plug-in estimator fits outcomes against own treatment and assignment-derived neighborhood features, evaluates the fitted response under all-treatment and all-control assignments, and subtracts the predictions. This makes the target explicit, but transports a fitted relationship outside much of its observed feature support.

Prior work establishes regression adjustments for this problem and demonstrates the usefulness of appropriate interference features (Chin, 2019; Han and Ugander, 2023). Exposure-weighted alternatives trade modeling assumptions for exposure probabilities and dependence conditions (Aronow and Samii, 2017; Ugander et al., 2013). These results do not imply that a low training error, a large sample, or a correctly included feature is individually sufficient for accurate GATE estimation. In particular, dense networks can concentrate neighbor treatment fractions around the allocation probability, making an endpoint contrast weakly informed even when outcomes are easy to predict locally.

This study asks a narrower, experimentally answerable question: *for fixed-feature linear regressions, which failure mechanisms determine endpoint error, and which apparently favorable diagnostics can be misleading?* We separate three issues: endpoint transport bias, contrast-specific noise

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amplification, and uncertainty calibration. Availability—whether an estimator can be computed at all—is reported separately. Our primary evidence is a controlled degree/size sweep, prespecified functional misspecifications, graph-design comparisons, and two public topologies with synthetic outcomes. The additive response and Gaussian error assumptions are deliberate controls, not descriptions of an actual platform.

The contributions are an integrated error audit: (i) a standard finite-sample linear-estimator decomposition used to distinguish modeling error from noise, with a lack-of-support counterexample; (ii) a complement-symmetry argument showing why quadratic misspecification can be unbiased in balanced designs while remaining high-risk; and (iii) fully reproducible, uncertainty-qualified comparisons across feature sets, designs, degrees, graph classes, and error dependence. The algebra is elementary; we do not claim that OLS variance formulas or support concerns are new. Nor do we claim exhaustive literature novelty: an explicit independence constraint restricts literature access to permitted originals (Appendix C). The value is the jointly controlled evidence and its failure boundaries, not a new universally best estimator.

2 Related work

Regression adjustment. Chin (2019) fits treatment-arm-specific response regressions using interference features and predicts the global endpoints. Linear results require exogeneity and appropriate rank conditions, with stronger assumptions for variance formulas. Our arm-specific fixed-feature regressions belong to this family; pooled additive regression is a lower-dimensional restriction. We do not reproduce the source paper's complete simulation program or nonlinear procedure. Han and Ugander (2023) generates graph/assignment covariates and selects them using ReFeX-LASSO, with a selection-aware graph-block bootstrap. The formal consistency result concerns the post variant under sparse linear representation and design assumptions; arbitrary feature generation is not unrestricted identification. We use known, fixed candidate maps to isolate their costs, not an implementation of ReFeX-LASSO or its bootstrap.

Exposure and experimental design. Aronow and Samii (2017) develops exposure-mapping-based randomized inference, requiring correctly mapped exposures and their probabilities. Ugander et al. (2013) shows how graph-cluster randomization can improve full-neighborhood exposure probabilities under graph/design conditions. We include exact marginal-probability full-exposure Horvitz–Thompson (HT) and fractional Hájek benchmarks, while distinguishing target misspecification, empty denominators, and rare tails. Our bounded-size greedy BFS partition is not their 3-net algorithm. Our contribution does not assert that this partition optimizes a design or that clustering uniformly reduces GATE risk.

3 Setting and error accounting

Let G be a fixed simple, undirected, unweighted graph with n nodes, no isolates, adjacency A, degrees d_i, and row-normalized adjacency P. For assignment $z \in \{0,1\}^n$, the neighbor fraction is $s = Pz$. Potential outcomes are

$$Y_i(z) = m_i(z) + \epsilon_i, \qquad \mathbb{E}[\epsilon \mid Z] = 0, \quad \mathrm{Var}(\epsilon \mid Z) = \Sigma. \tag{1}$$

The same error schedule is used at every potential assignment within an experiment. Thus the finite-population target is

$$T = \frac{1}{n}\sum_i \{Y_i(\mathbf{1}) - Y_i(0)\} = \frac{1}{n}\mathbf{1}^\top\{m(\mathbf{1}) - m(0)\}. \tag{2}$$

Assignments and error schedules are redrawn independently across Monte Carlo experiments. Reported coverage integrates both sources: it is not a randomization-only confidence guarantee for one fixed realized schedule.

Let $\Phi(z) \in \mathbb{R}^{n\times k}$ be fixed, outcome-independent features, with global contrast $c = n^{-1}\{\Phi(\mathbf{1}) - \Phi(0)\}^\top\mathbf{1}$. For a full-column-rank realized matrix $X = \Phi(Z)$,

$$\widehat{T} = c^\top(X^\top X)^{-1}X^\top Y = w^\top Y, \qquad w = X(X^\top X)^{-1}c. \tag{3}$$

Full column rank is our implementation rule, not a necessary condition for every identifiable contrast. We reject all singular fits rather than silently choosing a pseudoinverse prediction.

Proposition 1 (Conditional error accounting)**.** *Under (1), for any working coefficient β, let $r(z) = m(z) - \Phi(z)\beta$ and $\Delta r = n^{-1}\mathbf{1}^\top\{r(\mathbf{1}) - r(0)\}$. On a full-rank assignment,*

$$\widehat{T} - T = w^\top r(Z) - \Delta r + w^\top \epsilon, \tag{4}$$

$$b(Z) = \mathbb{E}[\widehat{T} - T \mid Z] = w^\top m(Z) - T,$$

$$\mathbb{E}[(\widehat{T} - T)^2 \mid Z] = b(Z)^2 + w^\top \Sigma w. \tag{5}$$

Under iid errors of variance σ^2, the variance is $\sigma^2 L(Z)$, where $L(Z) = \|w\|^2 = c^\top (X^\top X)^{-1} c$.

The proof substitutes $w^\top X = c^\top$ and takes first and second moments; no Gaussianity is needed. The full proof and ridge qualification appear in Appendix A. Averaging (5) over accepted assignments requires positive acceptance probability and gives $\mathbb{E}[b^2 + v \mid \text{accepted}]$, not $\mathbb{E}[b \mid \text{accepted}]^2 + \mathbb{E}[v \mid \text{accepted}]$. Variation of conditional bias is itself risk.

A useful but non-identifying sensitivity bound is

$$|b(Z)| \le \sqrt{L(Z)}\|r(Z)\| + |\Delta r|. \tag{6}$$

The endpoint residual Δr is unobserved. Even zero in-assignment signal residual cannot validate it. The decomposition is an oracle audit in simulations, where m and T are known; it is not an estimator of unknown causal bias from real experimental data.

3.1 A support obstruction

Proposition 2 (Indistinguishable endpoints)**.** *On K_n, even $n \ge 4$, randomize exactly $n/2$ treated nodes. The two response worlds $m_i^0(z) = z_i$ and $m_i^1(z) = z_i + \delta\mathbf{1}\{s_i(z) \ge .8\}$, $\delta > 0$, have identical observed distributions but GATEs 1 and $1 + \delta$. Any everywhere-defined estimator with finite second moment has maximum MSE over these worlds at least $\delta^2/4$.*

Indeed, observed $s_i \le n/[2(n-1)] \le 2/3$, so the extra term is never seen; at global treatment it equals δ. The squared-error lower bound follows by completing the square around the midpoint of the two targets. This result applies to the stated design and response class, not all Bernoulli experiments. OLS itself is singular for $(1, z, s)$ in this example.

Bernoulli assignment has positive but possibly negligible support. For a d-regular graph under independent probability $1/2$ assignment, a union bound gives

$$\Pr(\exists i : s_i \ge .8) \le \min\{1, n \Pr[\mathrm{Bin}(d, .5) \ge \lceil .8d \rceil]\}. \tag{7}$$

At $n = 1000, d = 128$, the computed upper bound is 9.67×10^{-10}. This is unconditional; it cannot be transferred unchanged to rank-filtered assignments. It explains why merely supplying the correct threshold feature can still leave no estimable coefficient.

3.2 Balanced symmetry can hide curvature

Proposition 3 (Quadratic cancellation)**.** *Let $\Phi(z) = (\mathbf{1}, x, z, s(z))$, with fixed x, and $m(z) = a\mathbf{1} + bx + z + s(z)^2$. Under a complement-symmetric assignment law, additive OLS GATE is unbiased conditional on full rank, provided this event has positive probability.*

Complementation sends (z, s) to $(1 - z, 1 - s)$ and the regression weight w to $-w$. The conditional bias is $w^\top s^2 - 1$ and changes sign under complementation. Full rank is invariant, so pairing assignments proves the claim (Appendix A.1). Both Bernoulli(.5) and independent cluster(.5) assignment are symmetric. This does not imply low variance, unbiasedness for general nonlinear effects, or validity at allocation probabilities .2/.8. A balanced quadratic benchmark is therefore an insufficient stress test of endpoint transport.

4 Experimental protocol

Graphs and assignments. The main suite uses $n = 1000$ random regular graphs of degree $8, 32, 128$, a four-block SBM with expected degree approximately 32 (80% within-block contribution),

BA graphs with 16 new edges per node, and WS graphs of degree 32 with rewiring probability .1. Each has three fixed graph seeds, 11/23/37. Public topologies are SNAP's combined ego-Facebook network (4039 nodes, 88234 edges) and Stanford3 (11586, 568309), with dataset snapshots hash-verified against the supplied versions (SNAP, 2013; Network Repository, 2026; Traud et al., 2012). They are distinct collections, not two samples of one graph population. We use topology only and synthesize all outcomes and covariates. Graph statistics are in Table 3 (Appendix B).

We compare independent node Bernoulli(.5) assignments with independently randomized graph blocks at probability .5. Blocks are deterministically constructed once by degree-ordered BFS, capped at 32 nodes. Between-block edges are retained. For each of 20 main graph instances and each design, we run 300 assignments, totaling 12,000 experiments. Prespecified sensitivities use 200 assignments per cell: regular graphs with $n = 500/2000$ at the same degrees, and allocation probabilities .2/.8 on regular32 and WS32, again three graph seeds. Main plus sensitivities total 24,000 assignments. RNG streams are reproducible but distinct across designs/suites; different designs are not analyzed as paired draws.

Response mechanisms. A fixed covariate x combines centered $\log(1 + d_i)$ with independent Gaussian node variation, standardized to mean zero and variance one. Every mechanism has $m_i(z) = .5x_i + z_i + h_i(z)$, with

$$h_i \in \{0,\ s_i,\ s_i^2,\ \mathbf{1}\{s_i \geq .8\},\ t_i,\ (d_i/\bar{d})s_i\}, \qquad t_i = \frac{(P^2 z)_i - (P^2)_{ii} z_i}{1 - (P^2)_{ii}}. \tag{8}$$

Here t is a normalized two-step exposure excluding returns to the same node, not a uniform average over distinct distance-two neighbors. All retained graphs have nonzero denominators. GATE is analytically 1 for no interference and 2 otherwise. A seventh mechanism repeats linear spillover with correlated noise. Default noise is iid $N(0, 1)$; correlated noise is

$$\epsilon = D(I + 2P)u, \quad u \sim N(0, I), \quad D_{ii} = (1 + 4/d_i)^{-1/2}. \tag{9}$$

Every marginal error variance is still one. Sharing assignments and u across mechanisms and estimators supports within-design paired comparisons; it does not create independent additional replications.

Estimators and ablations. DIM is the own-treatment mean difference. Add regresses on $(1, x, z, s)$. Lin fits separate-arm linear regressions on $(1, x, s)$, equivalently pooled $(1, x, z, zx, s, zs)$. Quad adds s^2 in each arm; TwoHop adds t to Lin; Count replaces s by $(d/\bar{d})s$ in Lin. This replacement avoids duplicate columns on regular graphs. Oracle uses only $(1, x, z, h)$ for the true mechanism (or $(1, x, z)$ without interference); it is an infeasible information benchmark, not a deployable method. RidgeQuad uses Quad with penalty one on raw-coordinate coefficients except the intercept. The penalty is fixed before the main study; no endpoint truth is used for tuning. These feature removals/substitutions isolate specification and variance costs without claiming data-adaptive model selection.

Hájek80 contrasts own-treated nodes with at least 80% treated neighbors against own-control nodes with at most 20% treated neighbors, weighted by exact marginal exposure probabilities. It returns unavailable if either weighted denominator is zero. HTfull uses full closed-neighborhood exposure and exact marginal propensities; it is unbiased for the one-hop mechanisms under positive propensities, but not generally for the two-hop mechanism. Under cluster assignment probabilities account for the clusters intersecting the closed neighborhood; fractional probabilities use a weighted Bernoulli convolution. These are target-qualified baselines, not automatically valid alternatives under arbitrary interference.

Metrics, uncertainty, and missingness. Primary risk is GATE RMSE, supplemented by bias, empirical SD, conditional b^2 and v, signal residual RMSE, availability, and interval coverage. OLS/ridge nominal intervals use the normal .975 quantile and residual variance times $\|w\|^2$; known-covariance intervals replace that variance by $w^\top \Sigma w$. Ridge intervals are a diagnostic, not justified unbiased regression inference. Residual-variance intervals are not implemented for DIM or exposure estimators. Raw outputs also include a generic known-noise interval diagnostic for their linear-in-outcome weights; it does not include randomization variability or confer valid GATE inference, and HT zero-weight intervals can be degenerate. Known-covariance coverage is only expected nominal with zero conditional contrast bias and positive variance.

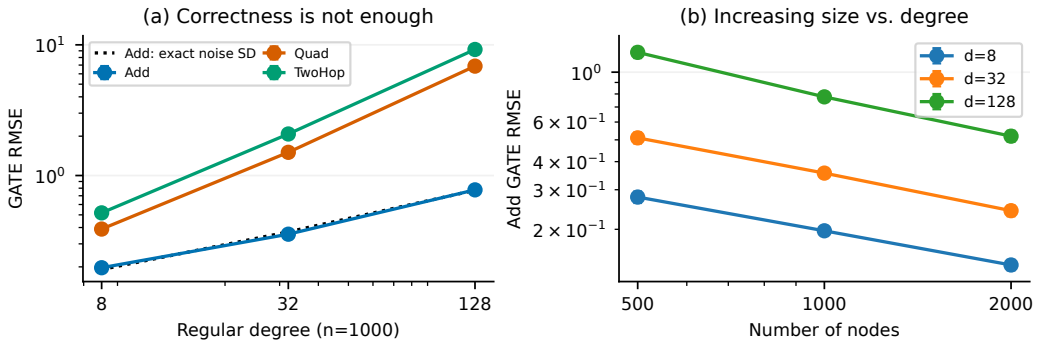


Figure 1: Correct linear outcome mechanism, Bernoulli(.5) assignment. Left: increasing degree at fixed $n = 1000$ inflates contrast variance; adding unnecessary quadratic or two-hop features further increases risk even though the models contain the truth. The dotted line is $\sqrt{\mathbb{E}[\|w\|^2]}$ for Add, not a fitted trend. Right: larger n reduces Add risk at each fixed degree. Points summarize three fixed graph seeds; bars are approximate 95% assignment/noise Monte Carlo intervals, not graph-population intervals.

We retain every attempt and report failures separately. Performance conditional on valid outputs must be read with availability; empty cells are not zero risk. For each fixed graph seed, MCSEs use independent assignments. Displayed synthetic aggregates describe a fixed three-graph mixture; mean-error and MSE MCSEs are stratified by graph seed, and RMSE uses the delta method. Tables show one MCSE; plotted error bars are approximate 95% Monte Carlo intervals. Coverage/availability use Wilson intervals; per-seed metrics and denominators are released. Successful-fit mixtures can reweight graphs when availability differs. These intervals do not quantify uncertainty over a graph population. No null-hypothesis or familywise superiority claims are made; all mechanisms and negative findings are retained. Rare-exposure heavy tails can defeat empirical MCSE diagnostics entirely.

5 Results

5.1 Contrast information, not fit alone, controls precision

Figure 1 holds graph size fixed while increasing regular degree. With correct additive linear features, Bernoulli RMSE increases from 0.197 at degree 8 to 0.777 at degree 128. Noise-integrated risk $\mathbb{E}[v]$ closely tracks empirical MSE; all 16 main Add/linear graph–design decomposition gaps lie within 1.96 MCSE of zero. This is a simulation check of the implementation, not evidence replacing Proposition 1.

A well-specified but unnecessarily flexible regression is not automatically better: Quad's degree-128 linear-mechanism RMSE is 6.872, despite its correctly included linear surface. Regular32's paired Quad-minus-Add MSE difference is 2.139 (MCSE .102, 900 common-valid assignments). For no interference, Bernoulli DIM RMSE is about .07 on the synthetic graphs, while introducing irrelevant neighbor features raises variance (full summaries released). Sample size helps at fixed degree, but the public-graph results cannot be inferred from degree alone: degree heterogeneity, graph structure, covariates, and size all change the realized contrast geometry.

Table 1 also separates two-hop misspecification from information scarcity. On Stanford3, adding t substantially improves the two-hop target. On high-degree regular graphs, even the correctly specified TwoHop estimator is extremely noisy. The Oracle two-hop estimator is also high-variance there (released summaries), so this is not merely an avoidable cost of own-arm interaction terms. A heuristic population-moment proxy for Add leverage is $4(1 + d)/n$, but it ignores cross-row dependence, centering, spectrum, and singularity; we use measured L rather than assert this as a universal graph rate.

Table 1: Bernoulli(.5) GATE RMSE (one MCSE in parentheses). Column labels specify estimator and true mechanism. Synthetic entries summarize three fixed graph seeds with 300 experiments each; each public entry has 300. All displayed fits are available. These are different response settings, not a single ranking task.

Graph	Add: linear	Quad: linear	Add: 2-hop	TwoHop: 2-hop	Count: count
Regular 8	0.197 (0.005)	0.389 (0.009)	1.036 (0.007)	0.518 (0.012)	0.197 (0.005)
Regular 32	0.356 (0.008)	1.505 (0.034)	1.070 (0.012)	2.079 (0.049)	0.358 (0.008)
Regular 128	0.777 (0.019)	6.872 (0.165)	1.233 (0.023)	9.216 (0.214)	0.777 (0.019)
SBM	0.355 (0.008)	1.499 (0.034)	1.025 (0.012)	1.588 (0.040)	0.328 (0.008)
BA	0.312 (0.008)	1.130 (0.028)	1.062 (0.010)	1.844 (0.040)	0.142 (0.004)
WS	0.362 (0.009)	1.693 (0.044)	0.613 (0.011)	0.598 (0.014)	0.363 (0.009)
ego-Facebook	0.108 (0.004)	0.200 (0.008)	0.790 (0.006)	0.245 (0.010)	0.048 (0.002)
Stanford3	0.063 (0.003)	0.106 (0.004)	0.968 (0.004)	0.224 (0.008)	0.033 (0.001)

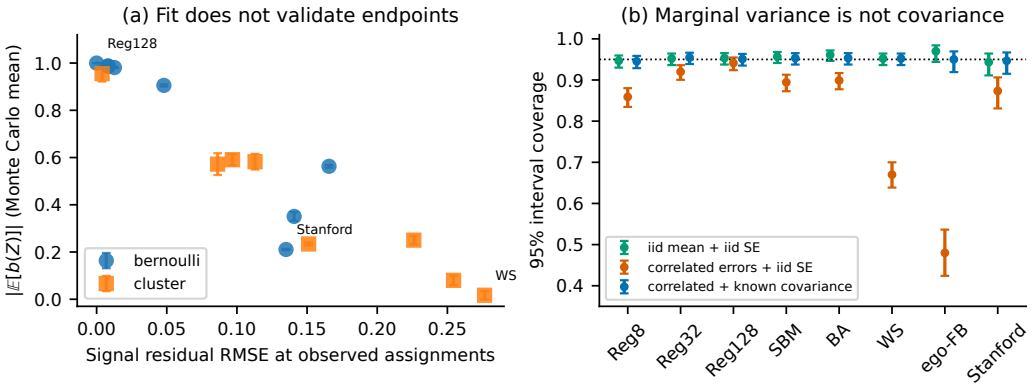


Figure 2: Two different failure modes. (a) Add under threshold interference: absolute mean conditional endpoint contrast bias versus mean in-assignment signal residual RMSE, for eight graph classes and two designs. Selected labels identify examples; vertical bars use MCSE of the conditional bias, not outcome residuals. (b) Add under correctly specified linear means, Bernoulli assignment: iid residual intervals versus known-covariance intervals. Correlated errors have unit marginal variance throughout. Coverage bars are Wilson 95% intervals across assignments/noise; the dotted line is .95.

5.2 Hidden endpoint effects and allocation sensitivity

For the degree-128 threshold mechanism, Add's mean signal residual RMSE is 0.000, yet its noise-integrated conditional bias is -1.000. The observed signal is effectively indistinguishable from no spillover; the missing global threshold effect remains one. Supplying the true threshold basis does not solve absent support: Oracle succeeds on 0/900 Bernoulli assignments and only 51/900 cluster assignments. The latter conditional RMSE must not be compared as if all 900 experiments produced estimates.

Quadratic spillover produces a different lesson. At probability .5, Add often has nearly zero *average* bias despite being misspecified, consistent with Proposition 3. On regular32, bias is $-.005$ (MCSE .012). At probabilities .2/.8 it becomes $-.558$ (MCSE .019) and .592 (MCSE .020), respectively. Figure 3 shows why evaluation only at balanced allocation can miss this vulnerability. Quad removes the functional misspecification but can pay a large variance cost; fixed ridge shrinks that cost while introducing its own endpoint bias. We do not select whichever estimator wins each scenario using known truth.

Degree-dependent spillover adds another model boundary. On Stanford3, using neighbor fraction for a degree-scaled-count mechanism yields Add bias -0.898, whereas the matched Count estimator has RMSE 0.033 (Table 1). The reverse substitution is not generally benign either. These results do not license Count as a uniformly better feature: it matches a different hypothesized response surface. On regular graphs count and fraction are scalar multiples and cannot be independently identified

Table 2: Threshold-mechanism diagnostics. B: Bernoulli; C: BFS-cluster assignment. Add bias is mean estimation error (MCSE); signal fit is residual RMSE against the noiseless observed mean. Valid/attempted counts refer to Oracle and Hájek80 (H80), not Add. Wilson intervals and per-seed denominators are in the released summaries.

Graph	Design	Add bias	Signal fit	Oracle valid	H80 valid
Regular 32	B	−0.994 (0.012)	0.008	210/900	11/900
Regular 32	C	−0.571 (0.015)	0.097	897/900	894/900
Regular 128	B	−0.956 (0.026)	0.000	0/900	0/900
Regular 128	C	−0.948 (0.030)	0.004	51/900	0/900
WS	B	−0.988 (0.012)	0.008	181/900	11/900
WS	C	0.018 (0.010)	0.277	900/900	900/900
ego-Facebook	B	−0.359 (0.011)	0.141	300/300	300/300
ego-Facebook	C	−0.072 (0.011)	0.254	300/300	300/300
Stanford3	B	−0.217 (0.004)	0.135	300/300	300/300
Stanford3	C	−0.239 (0.004)	0.151	300/300	300/300

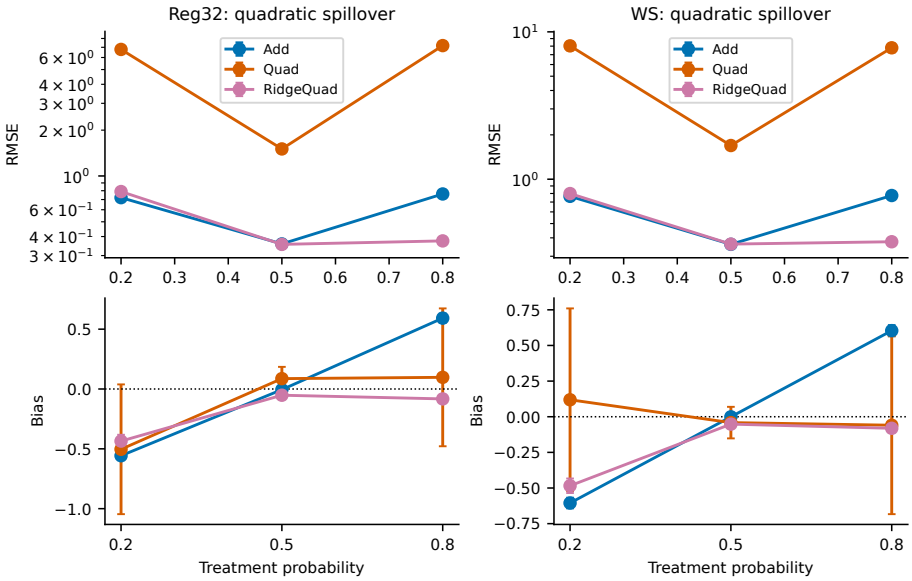


Figure 3: Prespecified allocation sensitivity under quadratic spillover, regular32 and WS32, Bernoulli assignment. Top: RMSE (log scale); bottom: bias. Add, Quad and fixed-penalty RidgeQuad exhibit different bias–variance tradeoffs. The .5 cells reuse the main suite; .2/.8 use 200 assignments per fixed graph seed, three seeds. Error bars are approximate 95% Monte Carlo intervals.

in a regression that includes both. Figure 4 gives additional feature-map comparisons, marking incomplete availability.

5.3 Clustering and uncertainty address different problems

Clustering improves exposure spread and some RMSEs, especially on WS, but does not uniformly dominate. BA/quadratic Add RMSE increases from .314 to .378 under our fixed cluster design, and the direction persists across all three graph seeds. Stanford3/threshold Add bias is not improved by this partition (Table 2). Clustering also changes the variability of conditional bias, so average unbiasedness need not imply calibrated intervals. We make no claims about alternative optimized partitions or maximum block sizes not studied.

With correlated errors, iid standard errors can be severely anticonservative despite the correct linear mean and unit marginal error variances. On ego-Facebook, Add coverage falls to 0.480 (Wilson 95% interval approximately .424–.536), while using known covariance gives 0.950 (approximately .919–.969). Figure 2 distinguishes this covariance failure from endpoint misspecification. Known-

covariance intervals remain non-nominal under biased response models; the benchmark is an oracle diagnostic, not an operational covariance estimator.

Rare exposure is not empirical unbiasedness testing. For regular32/Bernoulli/linear, HTfull returns zero in all 900 experiments: empirical bias is -2, and empirical bias MCSE is zero. Nevertheless the exact one-hop HT estimator is unbiased; the finite sample missed its enormous compensating rare tails. This is not evidence refuting the HT theorem, nor a usable precision guarantee. On Stanford3/Bernoulli, some full-exposure probabilities underflow floating point; all HTfull attempts are explicitly unavailable rather than clipped. Fractional Hájek has many empty exposure arms, while its nonempty estimates generally target an exposure-conditioned contrast rather than GATE for fractional nonlinear/multihop responses. Full status counts, zero-estimate fractions, and exposure diagnostics are released; unreliable rare-event RMSEs are not promoted as true-risk rankings.

6 Discussion, limitations, and conclusion

Regression GATE precision requires more than a rich predictor: the endpoint contrast must be informed by the assignment, the mean response must transport adequately, and covariance must be handled correctly. The audit suggests reporting contrast leverage, exposure range and rank availability alongside predictive diagnostics, explicitly varying plausible functional forms, and assessing allocation imbalance rather than relying only on balanced quadratic examples. These are diagnostics and sensitivity analyses, not data-only identification of the unobserved endpoint residual.

Scope is intentionally limited. The graph is known, static, unweighted, and undirected; outcome mechanisms are controlled functions with unit direct/spillover strengths. We do not study endogenous network formation, time dynamics, treatment-induced edges, real outcome heterogeneity, unknown networks, or graph-population inference. Covariance is known only for an oracle benchmark. Three synthetic graph realizations and two nonrepresentative public social topologies do not establish universal rates. Our arbitrary BFS partition and penalty are not tuned or optimal. Rank rejection can discard estimable singular contrasts; normal intervals and delta-method risk MCSEs are approximate, particularly fragile for rare-event tails. We did not implement data-adaptive ReFeX-LASSO, its block bootstrap, GNN estimators, or optimization-based designs; no comparative claim about those methods follows. The restrictive source policy prevents an exhaustive novelty assessment.

All findings are simulated; no real behavioral effect is estimated. Better network evaluation could reduce biased platform decisions, but false confidence in a model can also justify harmful global rollouts or obscure heterogeneous effects. These diagnostics should not replace domain-specific safety/fairness evaluation. Only public anonymized topology is used, with no profile attributes, reidentification, participants, or new platform data collection. Dataset license and attribution limitations are stated in Appendix C. This unattended research draft was developed with LLM assistance for hypotheses, proofs, code, analysis, and writing; execution artifacts substantiate numerical claims, but no human expert review or conference acceptance is implied. A responsible human author would still need to assess novelty, rights, and submission suitability.

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A Proofs and boundaries

Proposition 1. Since X has full column rank, $w^\top X = c^\top$. Write $m(z) = \Phi(z)\beta + r(z)$ for every assignment, including global endpoints. Then $T = c^\top\beta + \Delta r$, while $w^\top Y = c^\top\beta + w^\top r(Z) + w^\top\epsilon$. Subtraction gives (4). Conditional on Z, w is fixed, $\mathbb{E}[w^\top\epsilon \mid Z] = 0$ and its variance is $w^\top\Sigma w$. Expanding the square proves (5). With iid variance σ^2, $w^\top w = c^\top(X^\top X)^{-1}c$ by multiplying the definition of w. Cauchy–Schwarz and the triangle inequality prove (6). Conditioning on a positive-probability feature-only rank event preserves assignment–noise independence, but changes the assignment law. For a fixed graph the assignment space is finite; on accepted full-rank matrices all displayed moments exist under finite-variance noise.

For RidgeQuad, $\lambda = 1$, $D_0 = \mathrm{diag}(0, 1, \ldots, 1)$ and $w_\lambda = X(X^\top X + \lambda D_0)^{-1}c$. The penalized Gram matrix is positive definite because X has a nonzero intercept and every other coefficient is penalized. Directly, $b_\lambda = w_\lambda^\top m - T$, $v_\lambda = w_\lambda^\top\Sigma w_\lambda$ still imply MSE $b_\lambda^2 + v_\lambda$. However $w_\lambda^\top X = c^\top$ generally fails, so OLS residual cancellation is not asserted for ridge. A residual-variance interval need not cover a shrinkage-biased endpoint contrast.

For Gaussian noise with positive variance, the known-covariance interval centered at $w^\top Y$ has exact 95% conditional coverage if $b(Z) = 0$, using $q_{.975} = 1.959963984540054$. Full correct mean specification is sufficient, not necessary, for this contrast condition. Neither Gaussianity nor known covariance eliminates nonzero $b(Z)$. Zero-variance degenerate intervals need separate treatment; our reported regression cells have positive variance.

Proposition 2. A treated node in a balanced complete graph has neighbor fraction $(n/2-1)/(n-1)$ and a control node $(n/2)/(n-1)$. For even $n \geq 4$, both are below .8. Observed means are therefore identical in the two worlds at every supported assignment, as are observed distributions if the same assignment-independent noise law is added. At $\mathbf{1}$, the threshold is one at every node; at 0, it is zero. Thus targets differ by δ. For any real number a,

$$(a-1)^2 + (a-1-\delta)^2 = 2(a-1-\delta/2)^2 + \delta^2/2.$$

Taking expectation under the common observed law and dividing by two proves the maximum-risk bound. The triangle inequality similarly implies at least one absolute bias of $\delta/2$. This claim concerns estimators defined on all observations; an algorithm declining to estimate has availability, not an assigned MSE.

For (7), each node has d distinct, independently assigned neighbors, so its treated-neighbor count is binomial. A union bound over nodes suffices; exposures across nodes need not be independent. If R is a positive-probability rank-acceptance event, only the generic bound $\Pr(\exists i : s_i \geq .8 \mid R) \leq \min\{1, np_d/\Pr(R)\}$ follows from the unconditional tail p_d. The original binomial law need not survive filtering.

A.1 Complement proof

For Proposition 3, let $X = (\mathbf{1}, x, z, s)$. There is an invertible matrix B with $B^2 = I$, $X(\mathbf{1}-z) = XB$, and $B^\top c = -c$ for $c = (0, 0, 1, 1)^\top$. Explicitly,

$$B = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Thus the complement has the same rank, and

$$w(\mathbf{1}-z) = XB(B^\top X^\top XB)^{-1}c = X(X^\top X)^{-1}(B^\top)^{-1}c = -w(z).$$

The constraints are $w^\top\mathbf{1} = w^\top x = 0$ and $w^\top z = w^\top s = 1$. Consequently $b(z) = w^\top s^2 - 1$. At the complementary assignment,

$$b(\mathbf{1}-z) = -w^\top\{a\mathbf{1} + bx + (\mathbf{1}-z) + (\mathbf{1}-s)^2\} - 2 = 1 - w^\top s^2 = -b(z).$$

The positive-probability full-rank set is complement invariant. Pairing its finite assignment probabilities, which are equal by hypothesis, yields $\mathbb{E}[b \mid R] = 0$. Independent mean-zero noise does not change this expectation. Nonlinear threshold spillover, asymmetric allocation, and ridge penalty equivariance are not covered by this proposition. The result also does not establish correct intervals: $\mathrm{Var}(b(Z) \mid R)$ can be positive even when its mean is zero.

Correlated covariance. For a simple unweighted graph, $P_{ii} = 0$ and $\sum_j P_{ij}^2 = 1/d_i$, so (9) gives unit marginal variance. For any fixed w,

$$w^\top \Sigma w = \|(I + 2P)^\top Dw\|^2 = \|Dw + 2P^\top Dw\|^2.$$

Our code uses this sparse expression rather than forming a dense covariance matrix. The two-hop feature likewise uses two sparse matvecs and the diagonal return probability; it does not build a dense P^2.

B Full implementation and reporting details

Table 3: Main graph statistics for seed 11 (synthetic) or the fixed public graph. Full per-seed manifests are delivered. Blocks are capped at 32 but may be smaller; edge cut is the fraction of edges crossing the partition.

Graph	n	Mean d	Max d	Blocks	Edge cut
Regular 8	1000	8.00	8	54	0.746
Regular 32	1000	32.00	32	42	0.915
Regular 128	1000	128.00	128	33	0.955
SBM	1000	31.95	49	46	0.889
BA	1000	31.49	230	64	0.902
WS	1000	32.00	38	39	0.555
ego-Facebook	4039	43.69	1045	441	0.815
Stanford3	11586	98.10	1172	1653	0.964

Graphs are symmetrized, self-loops removed, and duplicate edges collapsed. Only the largest component would be kept; the reported public counts match the downloaded connected topologies. Stanford3 is the Facebook100 largest-component topology, not the SNAP ego-network union. The Stanford MatrixMarket archive has a single-percent header accepted by the installed SciPy reader. Original hashes, byte sizes, page snapshots, and source notes are in the delivered data manifest and access logs. The synthetic covariate is built using a seed distinct from assignment streams: centered $\log(1 + d_i)$, standardized when nonconstant (zero on regular graphs), plus .5 independent normal node variation, then standardized again. Thus regular graphs retain a nonconstant covariate without claiming real profile attributes.

The SBM has four nearly equal blocks with within probability $25.6/(n/4-1)$ and between probability $6.4/(3n/4)$. BFS roots are ordered by descending degree with stable ties; each unassigned root starts a queue, taking unassigned vertices until size 32, then moving to the next root. Leftover nodes start later blocks. This fixed partition is regenerated deterministically from the saved adjacency; no response or assignment is used in clustering. The block cap is not necessarily the realized average block size.

Design matrices use raw $s, s^2, t, (d/\bar{d})s$ coordinates. Eigenvalues of $X^\top X$ smaller than $10^{-10} \max(\lambda_{\max}, 1)$ count as numerically zero. OLS rejects deficient rank; positive-penalty ridge remains defined. Proposition 3 concerns exact mathematical rank: this raw-coordinate numerical cutoff need not be complement invariant. Add had no rejected fits in the reported suites, so no filtered pairs affect its reported cancellation evidence. The software records Gram condition numbers but does not select fits by condition number after full-rank acceptance. Every method recomputes its own endpoint feature contrast, including count normalization and treatment-feature interactions. Gaussian variance is estimated from residual sum of squares divided by $n - \mathrm{rank}(X)$; with ridge this is merely a diagnostic scale, not an effective-degrees-of-freedom adjustment.

For independent node assignment, full treatment/control propensities are p^{d_i+1} and $(1-p)^{d_i+1}$. Fractional treated exposure requires $Z_i = 1$ and neighbor count at least $\lceil .8d_i \rceil$; fractional control requires $Z_i = 0$ and count at most $\lfloor .2d_i \rfloor$. Cluster full-exposure propensities replace closed-neighborhood size by the number of distinct touching clusters. Fractional cluster propensities condition on own-cluster treatment and convolve the remaining cluster weights by their neighbor counts. The final 15-test suite includes exhaustive small-graph exposure-probability checks, endpoint truths, covariance orientation, rank handling, a balanced-graph counterexample, complementary weights/biases, attempted-record ratio uncertainty, and completeness-validation rejection tests. The initial seven tests preceded the

pilot; reporting tests were added after independent review. Numerical checks are not substituted for general proofs.

Each assignment uses `SeedSequence([20261102, suite, graphIndex, designIndex, rep])`, where suite codes are 0 pilot, 1 main, 2 size, and 3 allocation. Graph configuration order is defined in `run_experiments.py`; no hash-dependent RNG seeds are used. Main results exclude the five-replicate pilot, whose realized synthetic seed was only 11 despite a generic metadata list of all three graph seeds. Designs use separate streams, and mechanisms share noise only within an assignment. Scripts overwrite their own suite output directory; the README explicitly warns against accidental reruns over archived outputs.

For all-valid records of a fixed graph, MCSE of bias is $\mathrm{sd}(e)/\sqrt{R}$, of MSE is $\mathrm{sd}(e^2)/\sqrt{R}$, and of RMSE is MSE MCSE divided by $2\sqrt{\overline{e^2}}$. With failures, pooled performance is a ratio $\widehat{\theta} = \sum VH / \sum V$, where V indicates validity and H is the metric. Its target is the equal-attempt fixed-graph mixture conditioned on validity, which reweights graph seeds by success probability. Let $U_{jr} = V_{jr}(H_{jr} - \widehat{\theta})$, zero on invalid attempts. We use the attempted-record stratified delta variance $\widehat{\mathrm{SE}}^2 = \sum_j R_j \mathrm{var}_r(U_{jr})/N_{\mathrm{valid}}^2$, then apply the RMSE delta transform. This includes random accepted-count variation; it reduces to ordinary fixed-stratum mean uncertainty when all records are valid. Fewer than two successes gives unavailable MCSE; fewer than 30 successes is flagged as sparse. These local asymptotic diagnostics cannot certify unseen rare tails. This formula was corrected after self-review, with the original pre-main protocol and dated reporting amendment retained; experiment designs and outcomes were not changed. Per-seed conditional estimates and availability remain supplied. Wilson coverage intervals pool independent Bernoulli indicators as a fixed-mixture approximation; per-seed intervals are also available. Paired comparisons use all attempted keys and joint validity in the same ratio calculation, within a design only.

The direct accounting check uses $D_r = e_r^2 - b_r^2 - v_r$ on identical valid draws. We report its mean and MCSE. All three suites validate record uniqueness, finite values for successful estimates, expected assignment counts, and availability statuses before analysis. `summary.csv` has 2520 scenario/estimator cells and `per_seed.csv` has 7000; raw files contain 1,680,000 method–mechanism evaluations, not that many independent experiments. No rare-event trimming, pseudo-count smoothing, rank redraw, or replacement of failures by zero is performed.

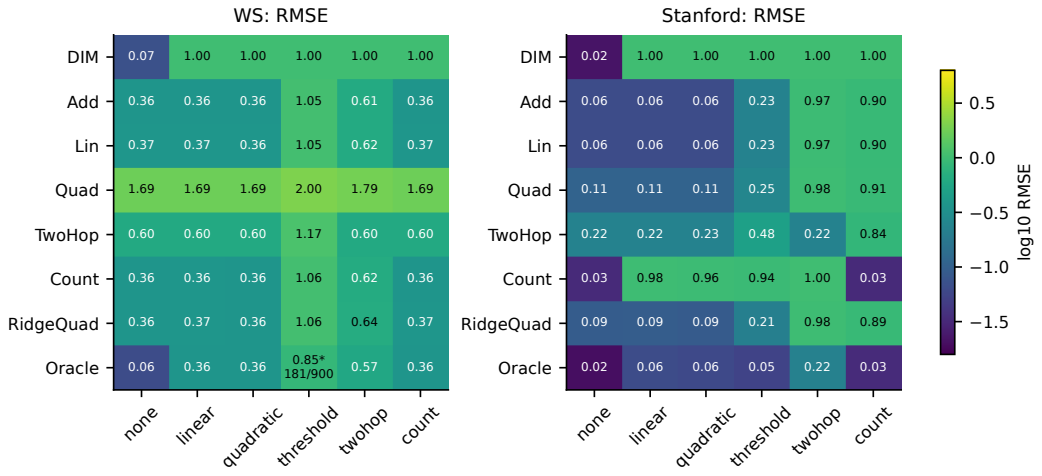


Figure 4: Additional Bernoulli(.5) RMSE comparisons on WS and Stanford3. Cells use measured simulation summaries, rounded for display; color is log10 RMSE. A star and valid/attempted counts mark incomplete cells; other WS cells use 900/900 and Stanford cells 300/300. These method-conditional risks are not common-subset comparisons. Oracle is a known-true-basis benchmark, not a learned method. Monte Carlo uncertainty is supplied in the CSVs; no significance rankings follow.

Reproducibility and compute. Use Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1, pandas 3.0.6, NetworkX 3.4.2, Matplotlib 3.11.2, pytest 8.3.5, and pypdf 5.9.0. Main simulation uses one

BLAS/OpenMP thread, no GPU, on a Linux x86-64 sandbox exposing 40 logical CPUs; the processor model is not exposed. Main wall time was 241.1 seconds; per-suite and analysis timings are delivered. These are measurements of this environment, not portability guarantees. Main peak memory was not instrumented; analysis peak RSS and environment metadata are recorded separately. The pilot and failed tooling attempts are logged. The entire research, including figures and tests, is substantially smaller than a week, but a new machine should allow additional time for package installation, graph downloads, and TeX.

The reproduction sequence is `fetch_data.py`; `pytest`; `run_experiments.py -suite main -reps 300`; corresponding size/allocation commands with 200; `analyze.py`; `plot_results.py`; `prepare_paper.py`; and `latexmk -pdf main.tex` in the paper directory. The root README supplies exact commands, runtime measurements, and every figure/table mapping. All empirical LaTeX tables/macros are generated from raw records. Analysis scripts, source code, and figure PDFs/PNGs are provided; no network access to unrelated literature is needed for reproduction.

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Answer: [N/A]

Justification: This simulation study uses previously public anonymized graph topology and no new participant interaction; no institutional approval is claimed.

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Answer: [Yes]

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